

# Harika Mahapatra

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## Research

### The Latent Truth: Knowledge–Behavior Gap in Logical Deduction

2024 – Present

*Mechanistic Interpretability & Activation Steering* (project page) — *preprint in progress*

- Identified a **knowledge–behavior gap**: LLMs internally encode correct logical truth (linear probes reach  $\sim 100\%$  by layer 8–16) yet behavioral output plateaus at  $\sim 65\%$ , across Gemma-2, Phi-2, and Mistral 7B.
- Demonstrated that even **explicit ground-truth symbolic reasoning** (Z3 solver traces) fails to close this gap, providing evidence suggesting the **final-layer unembedding process** acts as a decoding bottleneck.
- Applied **directional activation steering** as a causal intervention ( $\alpha=1.0$ ), recovering  $\sim 75\text{--}77\%$  of failures; random vector controls yield 0% recovery, confirming causal specificity; replicated across Gemma-2, Phi-2, and Mistral 7B.
- **Safety implication**: when a model's internal state systematically diverges from expressed output, it may internally represent correct reasoning while producing incorrect outputs, undermining behavioral evaluation as a proxy for alignment and complicating deception detection.

### HRLS: Interpretable Temporal Prediction for Clinical Decision Support

*NeurIPS WiML Workshop · IEEE Global AI Summit*

- Designed and developed a **residual hybrid architecture** (logistic regression + LSTM) framing deep learning as a corrector to a transparent baseline, preserving interpretability while capturing temporal dynamics.
- Proposed **Temporal Explanation Robustness (TER)**, a perturbation-based metric for evaluating explanation stability; achieved  $\rho=0.481$  ( $p<0.001$ ) on a 721-disease prediction task.
- Achieved **86.3% accuracy** competitive with Transformers and GRUs, while retaining a structurally interpretable baseline, with implications for **reliable model evaluation in safety-critical settings**.

### MAGF: Statistically-Grounded Attention for Alzheimer's Detection

*AJCAI (AIML-CEB) · IEEE Global AI Summit*

- Proposed and led development of a **parameter-free attention mechanism** grounded in the coefficient of variation, replacing learned attention weights with statistically principled reliability scores.
- Achieved **89.7% accuracy and 0.92 AUC** on the ADNI cohort (N=1,641), outperforming standard multi-head attention by 11.3% absolute accuracy.

## Education

### New York University, Courant Institute of Mathematical Sciences

Incoming Fall 2026

*M.S. in Computer Science*

### Sree Vidyanikethan Engineering College

2020 – 2024

*B.Tech in Computer Science and Engineering (Minor: Data Science)*

- CGPA: 8.9/10

## Experience

### Accenture

Jul 2024 – Present

*Software Engineer*

*Bangalore, India*

- Built data ingestion and processing pipelines for large-scale energy grid systems, ensuring reliability and integrity of time-series data used in downstream forecasting.
- Optimized distributed batch workflows through profiling and bottleneck analysis, reducing runtime by **20%**.
- Contributed to backend redesign during cloud migration, enforcing data integrity and standardizing transformations across systems.

## Activities & Service

### Technical Reviewer

2025

*EurIPS (endorsed by NeurIPS)*

- Reviewed submissions in machine learning for health, focusing on model reliability, evaluation methodology, and interpretability claims.

### Meta Hackathon

2022

*Rank 131 / 13,000+ participants (Top 1%)*

## Technical Skills

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**Languages:** Python, Java, SQL, Bash

**ML / Tools:** PyTorch, Hugging Face Transformers, Scikit-learn, TensorFlow, Docker, AWS

**Research Areas:** Mechanistic Interpretability, Activation Steering, LLM Reasoning, Model Reliability, Evaluation Methodology